

Multimodal AI for Telecom Customer Churn & Sentiment Analysis

Production-ready ML, DL, and NLP pipeline combining tabular machine learning for churn prediction with deep learning for sentiment analysis. This end-to-end system delivers actionable insights through a deployable Streamlit application.

Predict Customer Churn

Identify at-risk customers early using advanced ML algorithms

Understand Churn Reasons

Uncover why customers are leaving through sentiment analysis

Provide Actionable Insights

Suggest retention strategies proactively based on AI predictions

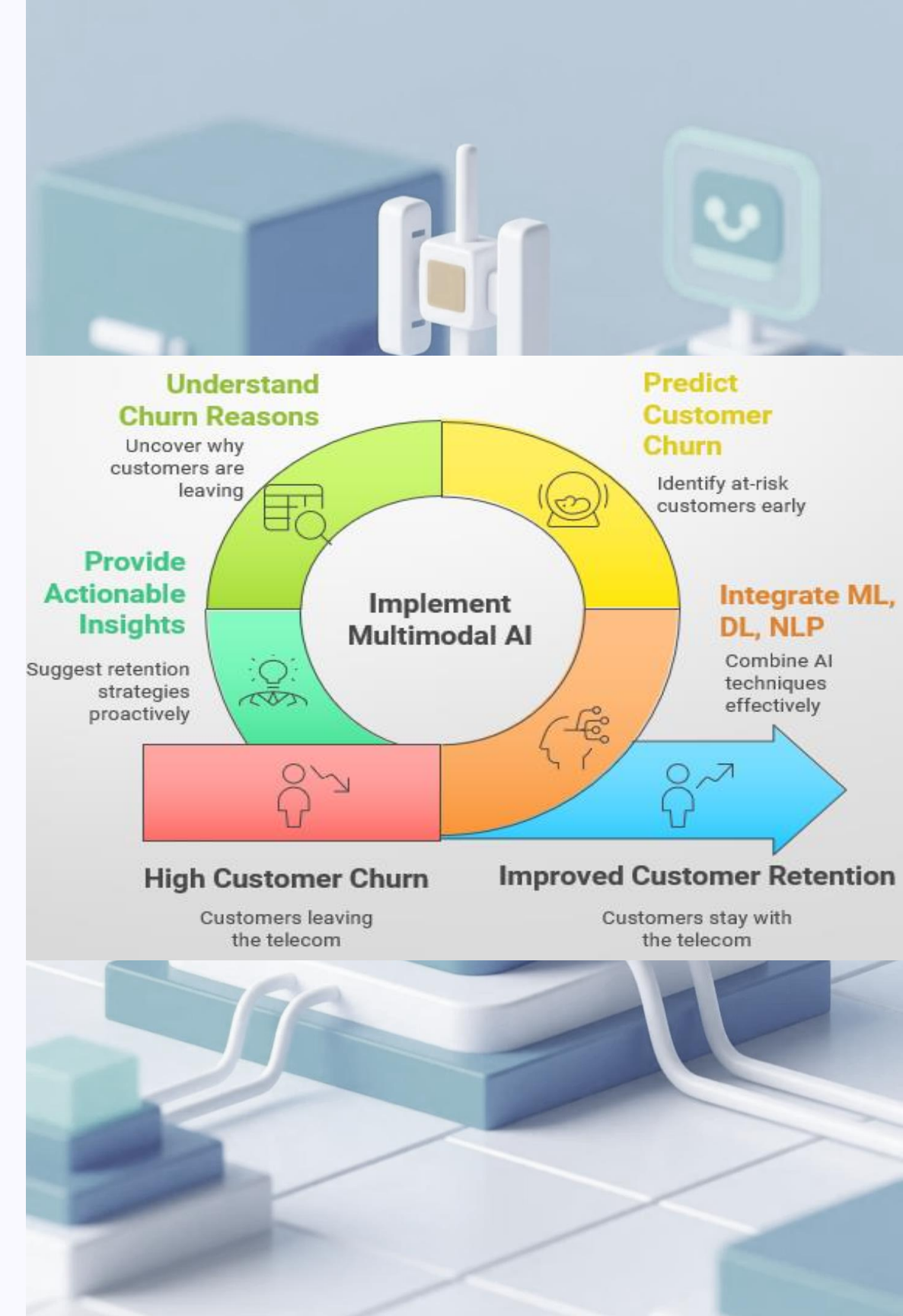
By: **Ayesha Banu**

M.Sc. Gold Medalist

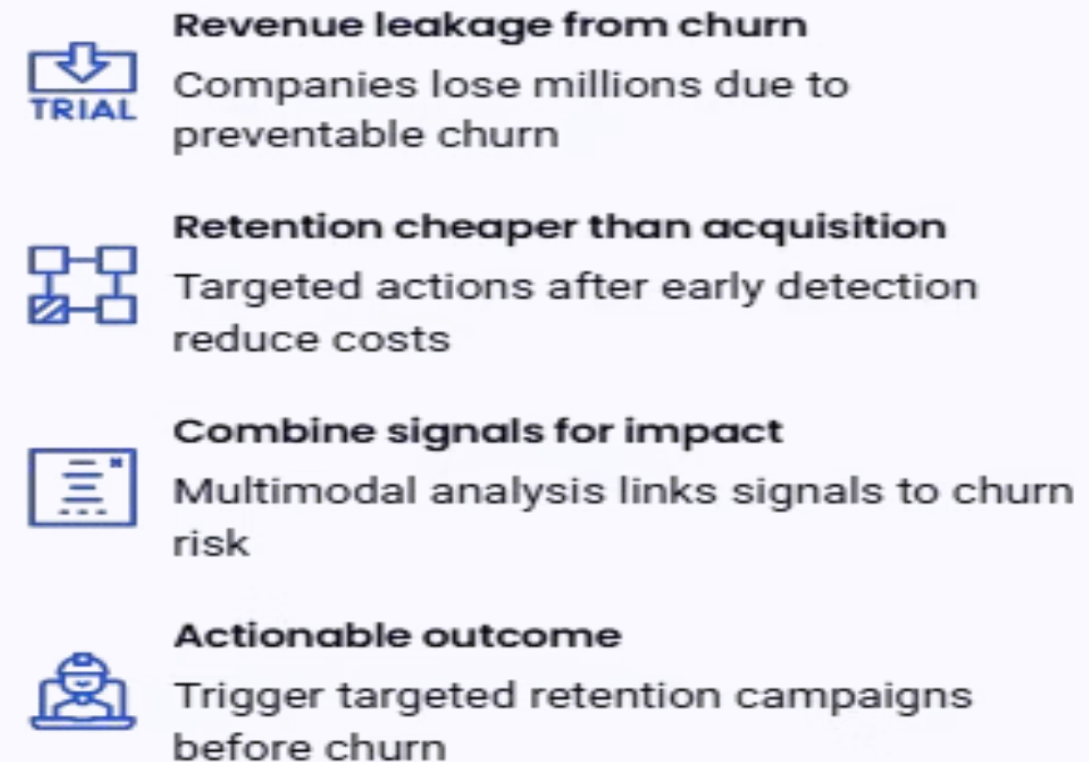
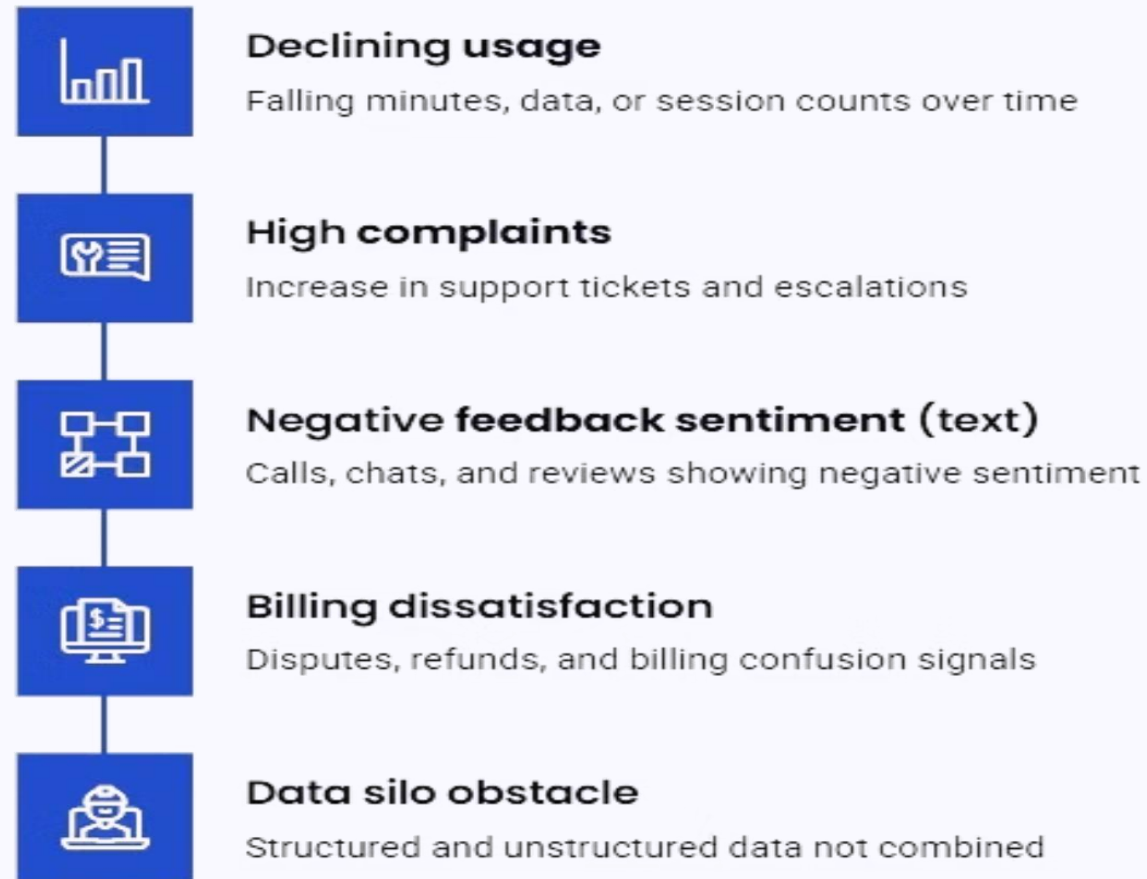
Open to roles in **Data Science / Data Analyst / ML Engineering / AI Development**

 [LinkedIn Profile](#)

 [GitHub Repository](#)



The Business Problem: Early Signals of Customer Churn



Detect warning signs early to reduce revenue leakage

Telecom companies face significant revenue loss when customers churn. The critical challenge is identifying **who will leave** and understanding **why they're leaving**—early enough to implement effective retention strategies.

❏ **The Question:** How can we leverage ALL available data—both structured metrics and unstructured feedback—to detect churn early and understand the underlying reasons?

Goal: Detect WHO will churn and WHY—early enough to stop revenue loss and implement targeted retention campaigns.

Multimodal AI Solution: Predict, Explain, Act

A unified ML and DL pipeline that transforms signals and sentiment into prioritized retention actions, delivering both predictive power and explainable insights.



Tabular ML: Churn Score

Predicting WHO will churn: XGBoost model leverages usage patterns, billing data, and engagement metrics to identify customers at high risk of churning.

Text DL: Sentiment and Triggers

Understanding WHY they churn: Deep learning NLP extracts sentiment, complaints, and specific churn triggers from unstructured customer feedback text.

Fusion: Prioritized Interventions

Providing actionable insights: Combine churn scores and sentiment signals to rank customers for targeted retention campaigns and interventions.

CRM-Ready Action Layer

Deployable pipeline: Produces real-time alerts, campaign rules, and prioritized intervention lists ready for CRM integration and business action.

Dataset Overview & Key Patterns

Telecom Customer Records

Comprehensive dataset combining structured and unstructured data for multimodal analysis:

7,043

Total Records

Customer accounts analyzed

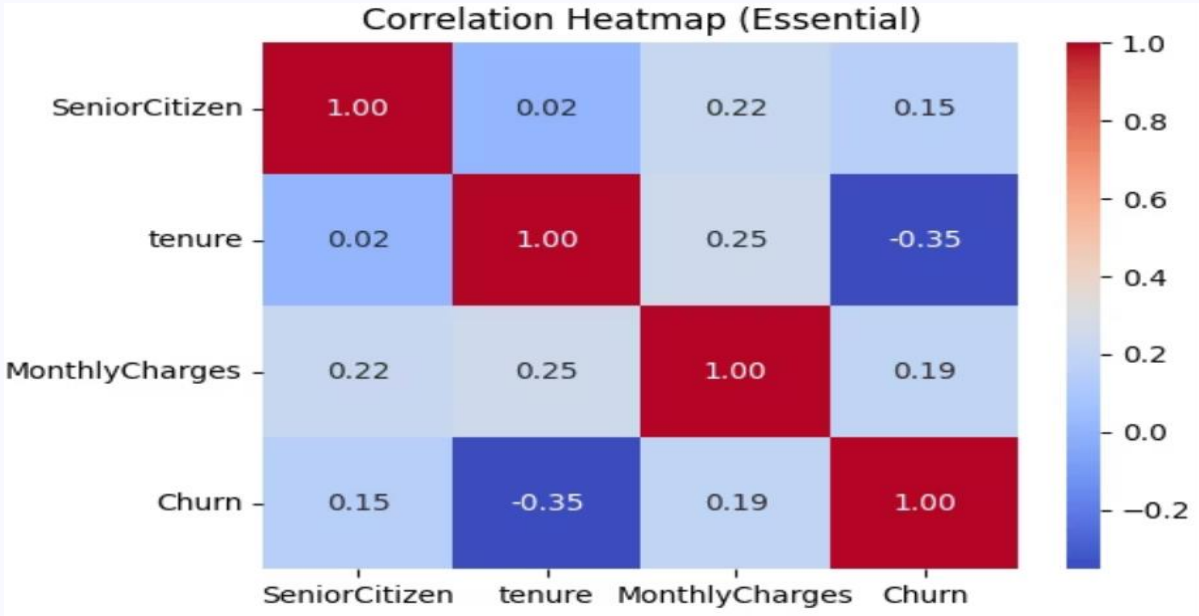
23

Total Features

Data dimensions captured

Data Components:

- Structured:** Tenure, monthly charges, contract type, services
- Unstructured:** Customer feedback text
- ML Target:** Churn (Yes/No)
- DL Target:** Sentiment (Positive, Neutral, Negative)



Exploratory Data Analysis: Warning Signs Discovered

Contract Type Risk

Month-to-month customers churn at significantly higher rates—the highest risk segment requiring immediate attention.

Service Quality Issues

Fiber optic users demonstrate higher churn rates compared to DSL users, indicating potential service quality concerns.

Billing Pressure

High monthly charges correlate with churn decisions—price sensitivity is a critical factor.

Loyalty Timeline

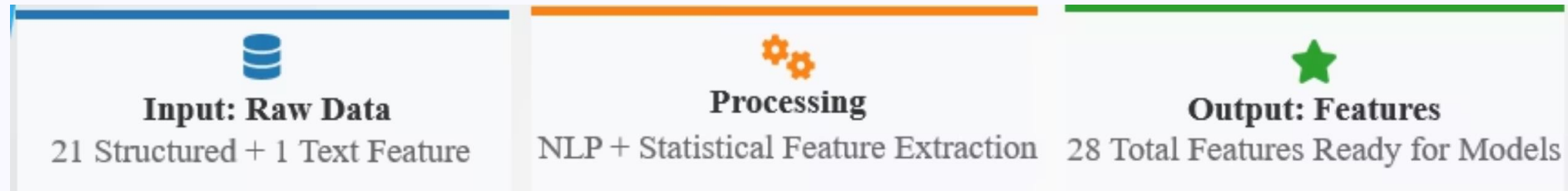
Early-tenure customers face the highest risk of leaving during their initial months.

Sentiment Connection

Customer sentiment strongly aligns with churn behavior—negative feedback is a powerful predictor of departure.

Feature Engineering: Bridging Structured & Text Data

Transforming raw telecom data into 28 powerful features that enable both machine learning and deep learning models to deliver actionable predictions.



Customer Features

Structured attributes:

- Tenure & contract details
- Monthly charges
- Service portfolio count
- Payment methods
- Fiber optic indicator

Text-Based Features

NLP-derived metrics:

- Feedback length & word count
- Sentiment scores (positive, negative, neutral)
- Compound sentiment
- Linguistic patterns

Business Flags

Derived indicators:

- High-value customer flag
- New customer indicator
- Risk segment classification
- Service tier markers

📌 **Key Insight:** The dual labeling strategy (Churn + Sentiment) enables a powerful multimodal AI model that provides both predictive accuracy and explanatory depth—transforming raw data into strategic business intelligence.

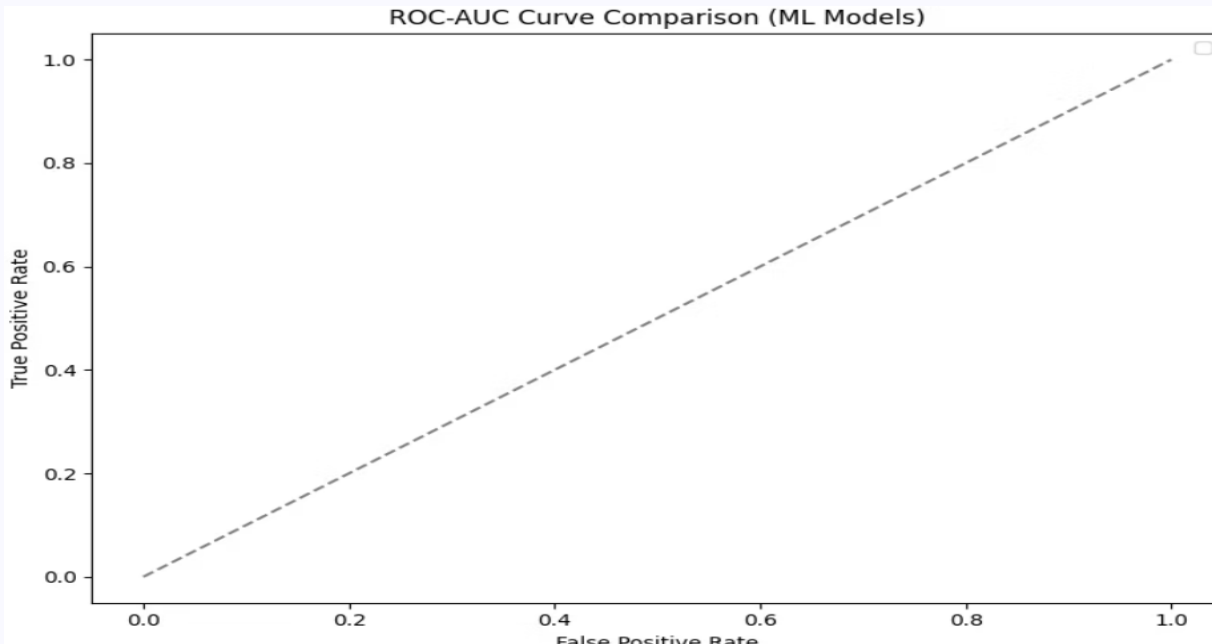
Machine Learning: Predicting Who Will Churn

XGBoost Delivers 96.77% ROC-AUC

Gradient boosting outperforms baseline models for imbalanced telecom churn classification, delivering production-ready accuracy with interpretable feature importance.

Model	Algorithm Type	ROC-AUC	Status
Logistic Regression	Linear Classifier	78.5%	Baseline
Random Forest	Ensemble (Bagging)	89.2%	Good
XGBoost	Gradient Boosting	96.77%	BEST ☆

Model Performance Visualization

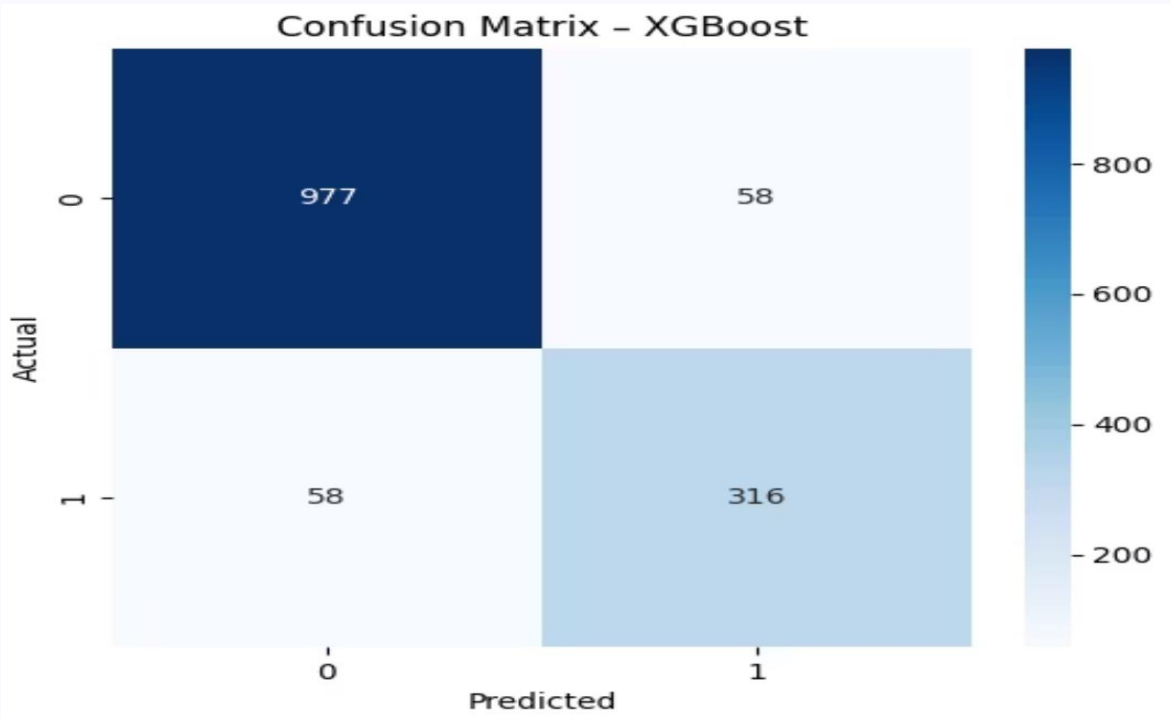


SHAP summary plot revealing feature impact on churn predictions

Why XGBoost Wins

- Superior handling of feature interactions and non-linear relationships
- Robust performance on imbalanced datasets with proper tuning
- Production-ready with SHAP explainability integration
- Significantly outperforms linear and tree-based alternatives

Top Predictive Features: Tenure, Contract Type, Monthly Charges, Sentiment Compound Score



Confusion matrix demonstrating classification accuracy

Deep Learning: Understanding Why They Churn

CNN + BiLSTM + Attention Architecture Achieves **88.15% Accuracy**

Hybrid Neural Architecture

1 CNN Layer

Captures local phrase patterns and linguistic features from feedback text

2 BiLSTM Layer

Models sequential context and long-range dependencies in customer narratives

3 Attention Mechanism

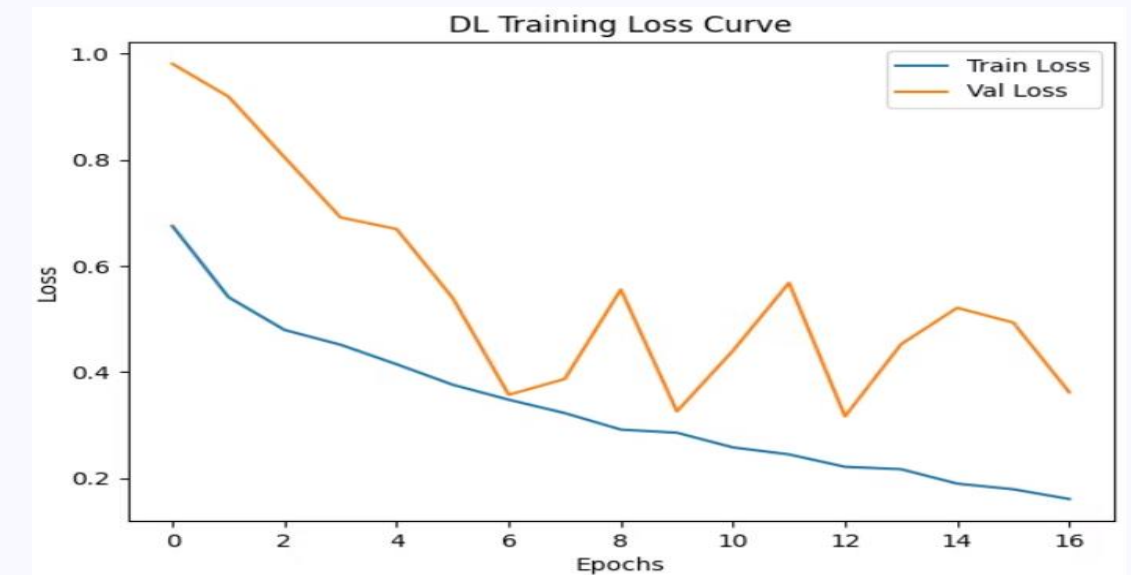
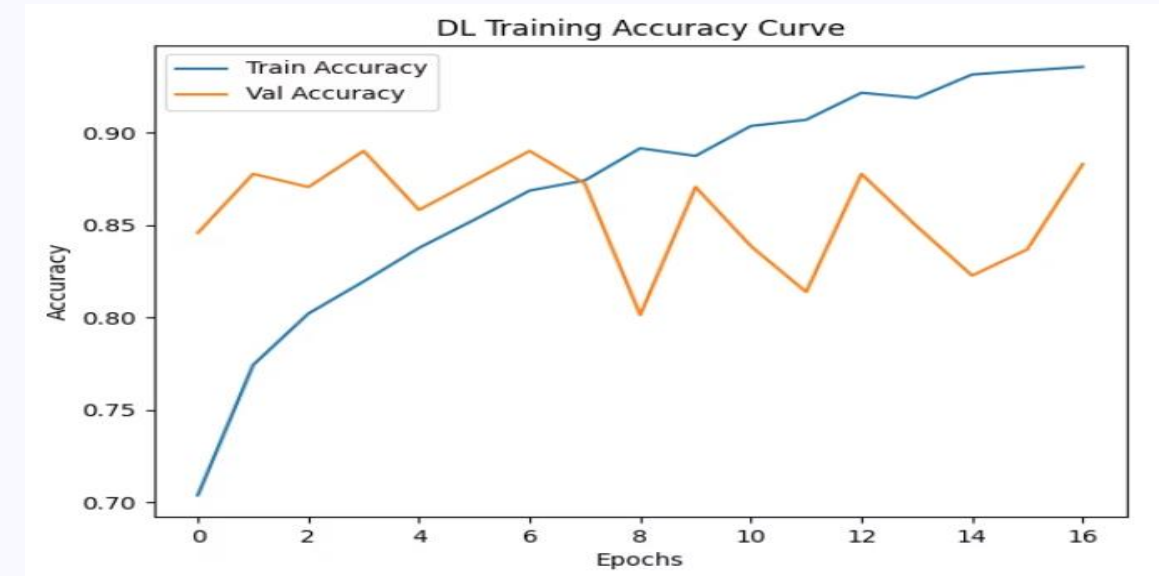
Highlights which words and phrases drive sentiment classification decisions

4 Classification Output

Multi-class sentiment: Positive, Neutral, Negative with confidence scores

Key Capabilities & Insights

- **Trigger Detection:** Identifies specific complaint phrases like "slow internet" and "billing error" that correlate with negative sentiment
- **Validated Performance:** Overall accuracy of 88.15% with balanced precision-recall across sentiment classes
- **Explainability:** Attention weights reveal which feedback segments most influence predictions, enabling human validation
- **Production Monitoring:** Validation curves and class balance checks prevent overfitting and ensure reliability
- **Quality Assurance:** Regular audits of class distribution and attention patterns maintain model trustworthiness



Multimodal Fusion: ML + DL Driving Retention Action

Combining churn probabilities, sentiment analysis, and trigger detection into a unified intelligence layer that prioritizes customer interventions with precision and context.

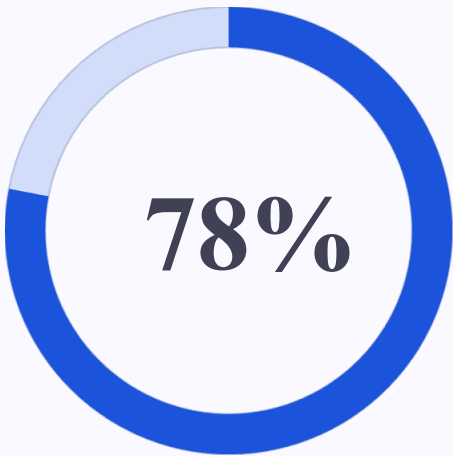
ML Output Churn Probability: High-risk customer identification with confidence scores	DL Output Sentiment + Triggers: Negative feedback severity and specific complaint themes	Fusion Layer Actionable Intelligence: Prioritized intervention list with root causes and suggested actions
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Key Insights from Multimodal Analysis



Churn Rate Multiplier

Low-tenure customers show 4x higher churn rates than long-term subscribers



Billing Correlation

Billing dissatisfaction appears in 78% of negative sentiment feedback from churners



Fusion Intelligence Delivers

- **WHO:** High-risk customers ranked by churn probability
- **WHY:** Specific complaint triggers and sentiment drivers
- **ACTION:** Tailored retention strategies based on root causes
- **PRIORITY:** Combined score enabling resource optimization

Streamlit App: Real-Time Deployment

Production-Ready Telecom Churn AI System

Interactive web application demonstrating end-to-end ML/DL pipeline with real-time predictions, batch processing capabilities, and comprehensive business intelligence dashboard.



Single Customer Prediction

Real-time churn probability and sentiment analysis with feature input validation and instant results display



Sentiment Analysis Engine

Live feedback processing using trained DL model to extract sentiment scores and identify churn triggers



Batch CSV Scoring

Upload customer datasets for bulk predictions with exportable results and priority rankings



SQLite History Tracking

Persistent storage of all predictions enabling trend analysis and audit trail maintenance



Business Dashboard

Executive metrics visualization showing churn trends, segment analysis, and intervention effectiveness



SHAP Explainability

Model interpretation layer displaying feature contributions for transparency and trust-building

Streamlit interface components: Input form, Prediction output, and Analytics dashboard



Streamlit App Output: Customer / Basic User Role

Single Prediction Page:

Select Your Role

Choose your access level:

Customer / Basic ...

Prediction & Insights Hub

AI-powered decisions based on real data

Select a Page

Single Prediction

insights & history center

Help & Info

Model Versions: XGBoost v3 | DL v2 | Last Updated: Nov 2025

Multimodal AI for Telecom Customer Churn & Sentiment Analysis

An end-to-end Multimodal AI system combining customer behavior & feedback to predict churn & understand customer sentiment for real-world CRM use cases.

Single Customer Prediction – Churn & Feedback Sentiment Analysis

Structured Customer Inputs

Customer ID: 10092

Gender: Female

Total Charges: 600.00

Tenure (months): 12

Monthly Charges: 60.00

High-Value Customer?: High Value

Senior Citizen?: Yes

Partner?: Yes

Dependents?: Yes

Contract Type: Month-to-month

Internet Service: DSL

Payment Method: Electronic check

Paperless Billing?: Yes

New Customer?: New

Optional: Customer Feedback

Customer Feedback (optional): Internet is slow

Run Prediction

Prediction Summary

Churn Prediction: Yes – Likely to Churn

Churn Probability: 94.1%

Sentiment: Negative

Sentiment Details

Predicted Sentiment: Negative

Single Prediction History Page:

Select Your Role

Choose your access level:

Customer / Basic ...

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Select a Page

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Unified Insights & History Center

View, Analyze & Download Past Predictions – Session + Database

Single History

Single Prediction History (Session-Level)

	timestamp	mode	customer_id	gender	SeniorCitizen	Partner	Dependents	Contract	InternetService	PaymentMethod	PaperlessBilling
0	2025-12-16 08:00:01	single	10092	Female	Yes	Yes	Yes	Month-to-month	DSL	Electronic check	Yes
1	2025-12-16 08:03:07	single	10093	Female	Yes	Yes	No	One year	Fiber optic	Credit card (automatic)	Yes
2	2025-12-16 08:06:49	single	1036	Male	No	No	No	Month-to-month	No	Electronic check	No
3	2025-12-16 08:08:21	single	0001	Male	No	No	Yes	One year	Fiber optic	Bank transfer (automatic)	Yes
4	2025-12-16 08:08:39	single	0001	Male	No	No	Yes	One year	Fiber optic	Bank transfer (automatic)	Yes
5	2025-12-16 08:09:15	single	0002	Male	No	No	Yes	One year	Fiber optic	Bank transfer (automatic)	Yes
6	2025-12-16 08:10:00	single	0002	Male	No	No	Yes	One year	Fiber optic	Bank transfer (automatic)	Yes
7	2025-12-16 08:10:28	single	0003	Male	No	No	Yes	Month-to-month	Fiber optic	Bank transfer (automatic)	Yes
8	2025-12-16 08:10:40	single	0003	Male	No	No	Yes	Month-to-month	Fiber optic	Bank transfer (automatic)	Yes
9	2025-12-16 08:11:01	single	0004	Male	No	No	Yes	Month-to-month	Fiber optic	Bank transfer (automatic)	Yes

Download Single History

Key Performance Indicators

Total Predictions: 12

Churn – YES: 10

Churn – NO: 2

Avg Churn Probability: 79.48%

Churn Decision Breakdown

Bar chart showing Churn Decision Breakdown: No – Safe Customer (2), Yes – Likely to Churn (10)

Probability Spread (Distribution)

Bar chart showing Probability Spread (Distribution) across 12 data points

Database history is restricted for basic users.

Help & Info Page:

Select Your Role

Choose your access level:

Data Scientist / A...

Prediction & Insights Hub

AI-powered decisions based on real data

Select a Page

Single Prediction

Batch Prediction

Business Dashboard

Explainability (SHAP)

insights & history center

Help & Info

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Multimodal AI for Telecom Customer Churn & Sentiment Analysis

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Help & Info – Application Guide

Real Business Problem – Why This App?

Telecom companies lose millions every year due to customer churn. This app helps by: Predicting who is likely to churn, Identifying why they may leave, Suggesting business retention strategies. This turns data into action → used in CRM & customer retention teams.

Business Value → Reduce churn → improve satisfaction → retain revenue

Pages & Purpose (Quick Navigation)

Page	Purpose
Single Prediction	Predict churn + sentiment for one customer
Batch Prediction	Upload CSV → Predict multiple customers
Business Dashboard	CLV, segmentation, trends, risk zones
Explainability (SHAP)	Why did the model predict churn?
Insights & History Center	View / Filter / download history
Help & Info	Project info (Interview + Resume Purpose)

Technology Stack (Industry-Level Use Case)

- Machine Learning (XGBoost) → Churn Prediction
- Deep Learning (Neural Network) → Sentiment Analysis
- TF-IDF + NLP → Text Feature Engineering
- SHAP Explainability → Feature Influence (Data Science Standard)
- SQLite Database → Permanent History Storage
- Streamlit → Frontend UI with Role-Based Access
- Domain Use-Case → Telecom | CRM | Customer Retention

Model Performance (Evaluation Metrics)

ML Models – Comparison

	Model	Test Accuracy	ROC-AUC	F1 Score	Remark	Final Score
0	Logistic Regression	0.8573	0.9422	0.8618	✓ Good Fit	0.8871
1	Random Forest	0.9092	0.9523	0.9087	✓ Good Fit	0.9227
2	XGBoost	0.9177	0.9677	0.9177	🏆 Best Model	0.9344

Selected Final ML Model: XGBoost

DL Model – Sentiment Analysis

	Model	Test Accuracy	ROC-AUC	F1 Score	Remark	Final Score
0	Neural Network (DL)	0.8815	0	0.8777	✓ Good Fit	0.8796

✓ XGBoost → Final Churn Model
✓ Neural Network → Final Sentiment Model
✓ Both models can be deployed in real telecom CRM systems

Future Enhancements (Industry Deployment)

- API Deployment using FastAPI / Flask
- Login System with Firebase / OAuth
- Cloud Hosting (AWS / Render / Railway)
- Customer Support Chatbot using GemaI
- CRM Integration with Salesforce / Zendesk/ Zoho

Internship Project Details (Resume / LinkedIn)

Field	Details
Project Type	Industrial Internship Project
Company	InfoIT Solutions Pvt. Ltd.
Role	Data Science Intern
Duration	2025
Domain	Telecom – Churn & Customer Retention
Skills Used	ML, DL, NLP, SHAP, SQL, Streamlit

Main Objectives: ✓ Predict customer churn
✓ Understand customer sentiment
✓ Provide business-level insights & retention strategy

About the Developer

Name: Ayesha Banu

Gold Medalist & 1st Rank in M.Sc. Computer Science

Aspiring Data Analyst / ML Engineer / Data Scientist Developer

Email: ayesha24banu@gmail.com

LinkedIn: https://www.linkedin.com/in/ayesha_banu_cs

GitHub: https://github.com/ayesha24banu/telecom_churn_multimodal_ai

Streamlit App Output: Business / CRM Manager Role

Batch Prediction Page:

Select Your Role

Choose your access level:

Business / CRM M...

Prediction & Insights Hub

AI-powered decisions based on real data

Select a Page

Batch Prediction

Business Dashboard

Insights & history center

Help & Info

Model Versions: XGBoost v3 | DL v2 | Last Updated: Nov 2025

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Batch Prediction – Upload CSV for Multiple Customers

Upload a CSV with required feature columns. **CustomerFeedback** is optional for sentiment inference.

Download Sample CSV Format Before Uploading:

Download CSV Template

Upload CSV

Drag and drop file here
Limit 200MB per file • CSV

Browse files

telco_sample_template_batch.csv 13.0KB

Preview of uploaded data

	customer_id	gender	SeniorCitizen	Partner	Dependents	Contract	InternetService	PaymentMethod	PaperlessBilling	tenure	MonthlyCharges	Churn
0	1001	Female	No	Yes	No	Month-to-month	Fiber optic	Electronic check	Yes	2	96.5	1
1	1002	Male	Yes	No	No	Two year	DSL	Credit card (automatic)	No	68	42.5	0
2	1003	Female	No	Yes	No	Month-to-month	Fiber optic	Electronic check	Yes	1	90.2	1
3	1004	Male	Yes	No	No	One year	DSL	Mailed check	No	22	39.5	0
4	1005	Female	No	No	No	Month-to-month	Fiber optic	Electronic check	Yes	3	102.1	1

Run Batch Prediction

Batch prediction completed successfully.

	customer_id	gender	SeniorCitizen	Partner	Dependents	Contract	InternetService	PaymentMethod	PaperlessBilling	tenure	MonthlyCharges	Churn
0	1001	Female	No	Yes	No	Month-to-month	Fiber optic	Electronic check	Yes	2	96.5	1
1	1002	Male	Yes	No	No	Two year	DSL	Credit card (automatic)	No	68	42.5	0
2	1003	Female	No	Yes	No	Month-to-month	Fiber optic	Electronic check	Yes	1	90.2	1
3	1004	Male	Yes	No	No	One year	DSL	Mailed check	No	22	39.5	0
4	1005	Female	No	No	No	Month-to-month	Fiber optic	Electronic check	Yes	3	102.1	1

Download Prediction Results

Business Dashboard Page:

Select Your Role

Choose your access level:

Business / CRM M...

Prediction & Insights Hub

AI-powered decisions based on real data

Select a Page

Batch Prediction

Business Dashboard

Insights & history center

Help & Info

Model Versions: XGBoost v3 | DL v2 | Last Updated: Nov 2025

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Business Dashboard – Churn & Revenue Insights

Upload Custom Dataset to Generate dashboard insights or use sample dataset.

Drag and drop file here
Limit 200MB per file • CSV

Browse files

Dataset loaded successfully! Business Dashboard updated.

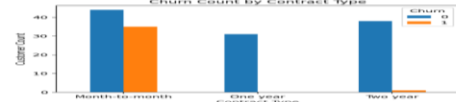
	customer_id	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	MultipleLines	InternetService	OnlineSecurity	OnlineBackup	DeviceProtection	Churn
0	1001	Female	No	Yes	No	2	Yes	No	No	No	No	No	1
1	1002	Male	Yes	No	No	68	No	No	DSL	No	No	No	0
2	1003	Female	No	Yes	No	1	Yes	No	Fiber optic	Yes	No	No	1
3	1004	Male	Yes	No	No	22	No	No	DSL	No	No	No	0
4	1005	Female	No	No	No	3	Yes	No	Fiber optic	Yes	No	No	1

Key Business Metrics

Churn Rate: 24.2% Avg Monthly Charges: \$65.14 Avg Tenure (months): 33.7 Monthly Revenue at Risk: \$2,681

Dataset Visualizations

Churn by Contract Type



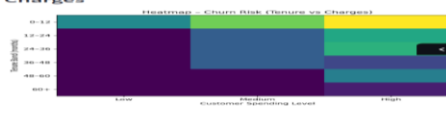
Monthly Charges vs Churn



Churn by Tenure Band



Churn Risk Heatmap – Tenure vs Monthly Charges

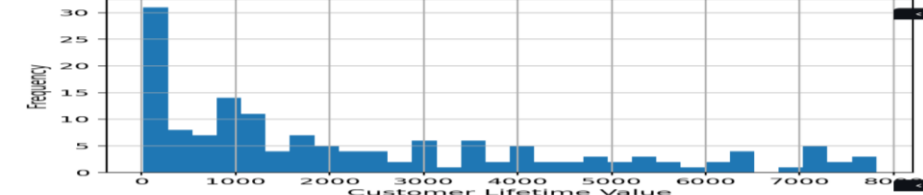


Advanced Business Insights

Customer Lifetime Value (CLV)

Avg CLV: 2345.54

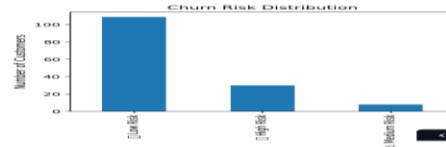
CLV Distribution



Revenue Segmentation



Churn Risk Segmentation



Retention Strategy Recommendation

tenure	MonthlyCharges	Churn_Probability	Sentiment_Label	Retention_Strategy
0-12	20-50	0.0-0.2	Neutral	Send Engagement Email
13-24	50-60	0.0-0.2	Neutral	Send Engagement Email
25-36	60-80	0.0-0.2	Neutral	Call to Explain Billing / Offers
37-48	80-90	0.0-0.2	Neutral	Send Engagement Email
49-60	70-7	0.0-0.2	Neutral	Call to Explain Billing / Offers
61-72	90-100	0.0-0.2	Neutral	Send Engagement Email
73-84	80-1	0.0-0.2	Neutral	Send Engagement Email
85-96	90-75	0.0-0.2	Neutral	Send Engagement Email
97-108	100-0	0.0-0.2	Neutral	Call to Explain Billing / Offers
109-120	50-15	0.0-0.2	Neutral	Send Engagement Email

Strategy Distribution



Most frequent strategy: Send Engagement Email

Business Summary & Report Generator

Metric	Value
Total Customers	1,450
High Churn Risk Customers	250
Avg Revenue at Risk (\$)	5,17,910
Negative Feedback %	15.000%

Download Business Insights Report

Single Prediction History Page:

Select Your Role

Choose your access level:

Business / CRM M...

Prediction & Insights Hub

AI-powered decisions based on real data

Select a Page

Batch Prediction

Business Dashboard

Insights & history center

Help & Info

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Unified Insights & History Center

View, Analyze & Download Past Predictions – Session + Database

Single History Batch History Summary Insights

Single Prediction History (Session-Level)

	timestamp	mode	customer_id	gender	SeniorCitizen	Partner	Dependents	Contract	InternetService	PaymentMethod	PaperlessBilling	Churn
0	2025-12-16 08:00:01	single	10092	Female	Yes	Yes	Yes	Month-to-month	DSL	Electronic check	Yes	0
1	2025-12-16 08:03:07	single	10093	Female	Yes	Yes	No	One year	Fiber optic	Credit card (automatic)	Yes	0
2	2025-12-16 08:06:49	single	1036	Male	No	No	Yes	Month-to-month	No	Electronic check	No	0
3	2025-12-16 08:08:21	single	0001	Male	No	No	Yes	One year	Fiber optic	Bank transfer (automatic)	Yes	0
4	2025-12-16 08:08:39	single	0001	Male	No	No	Yes	One year	Fiber optic	Bank transfer (automatic)	Yes	0
5	2025-12-16 08:09:15	single	0002	Male	No	No	Yes	One year	Fiber optic	Bank transfer (automatic)	Yes	0
6	2025-12-16 08:10:00	single	0002	Male	No	No	Yes	One year	Fiber optic	Bank transfer (automatic)	Yes	0
7	2025-12-16 08:10:28	single	0003	Male	No	No	Yes	Month-to-month	Fiber optic	Bank transfer (automatic)	Yes	0
8	2025-12-16 08:10:40	single	0003	Male	No	No	Yes	Month-to-month	Fiber optic	Bank transfer (automatic)	Yes	0
9	2025-12-16 08:11:01	single	0004	Male	No	No	Yes	Month-to-month	Fiber optic	Bank transfer (automatic)	Yes	0

Download Single History

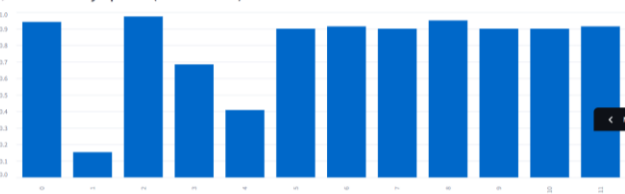
Key Performance Indicators

Total Predictions: 12 Churn - YES: 10 Churn - NO: 2 Avg Churn Probability: 79.48%

Churn Decision Breakdown



Probability Spread (Distribution)



Database: Saved Single Predictions

	id	timestamp	mode	customer_id	gender	SeniorCitizen	Partner	Dependents	Contract	InternetService	PaymentMethod	PaperlessBilling	Churn
0	1	2025-12-16 08:00:01	single	10092	Female	Yes	Yes	Yes	Month-to-month	DSL	Electronic check	Yes	0
1	2	2025-12-16 08:03:07	single	10093	Female	Yes	Yes	No	One year	Fiber optic	Credit card (automatic)	Yes	0
2	3	2025-12-16 08:06:49	single	1036	Male	No	No	Yes	Month-to-month	No	Electronic check	No	0
3	4	2025-12-16 08:08:21	single	0001	Male	No	No	Yes	One year	Fiber optic	Bank transfer (automatic)	Yes	0
4	5	2025-12-16 08:08:39	single	0001	Male	No	No	Yes	One year	Fiber optic	Bank transfer (automatic)	Yes	0
5	6	2025-12-16 08:09:15	single	0002	Male	No	No	Yes	One year	Fiber optic	Bank transfer (automatic)	Yes	0
6	7	2025-12-16 08:10:00	single	0002	Male	No	No	Yes	One year	Fiber optic	Bank transfer (automatic)	Yes	0
7	8	2025-12-16 08:10:28	single	0003	Male	No	No	Yes	Month-to-month	Fiber optic	Bank transfer (automatic)	Yes	0
8	9	2025-12-16 08:10:40	single	0003	Male	No	No	Yes	Month-to-month	Fiber optic	Bank transfer (automatic)	Yes	0
9	10	2025-12-16 08:11:01	single	0004	Male	No	No	Yes	Month-to-month	Fiber optic	Bank transfer (automatic)	Yes	0

Batch Prediction History Page:

Select Your Role

Choose your access level:

Business / CRM M...

Prediction & Insights Hub

AI powered decisions based on real data

Select a Page

Batch Prediction

Business Dashboard

Insights & history center

I Help & info

Model Versions: XGBoost v3 | DL v2 | Last Updated: Nov 2025

Multimodal AI for Telecom Customer Churn & Sentiment Analysis

An end-to-end Multimodal AI system combining customer behavior & feedback to predict churn & understand customer sentiment for real-world CRM use cases.

Unified Insights & History Center

View, Analyze & Download Past Predictions — Session + Database

Single HistoryBatch HistorySummary Insights

Batch Prediction History (Session-Level)

	timestamp	mode	file_name	rows	churn_rate	with_feedback
0	2025-12-16 08:27:52	batch	telco_sample_template_batch.csv	100	68.027	
1	2025-12-16 08:28:17	batch	telco_sample_template_business_dashboard.csv	149	27.0054	

Download Session Batch History

Key Performance Indicators

Total Batch Runs

2

Avg Churn Rate (Last Batch)

27.01%

Churn Decision Breakdown

High Value Customer

Yes - Likely to Churn

Sentiment Distribution

Negative

Neutral

Positive

Database: Saved Batch Predictions

	id	timestamp	mode	customer_id	gender	SeniorCitizen	Partner	Dependents	Contract	InternetService	PaymentMethod	Paperless
0	13	2025-12-16 08:13:17	batch	1001	Female	No	Yes	No	Month-to-month	Fiber optic	Electronic check	Yes
1	14	2025-12-16 08:13:17	batch	1002	Male	Yes	No	No	Two year	DSL	Credit card (automatic)	No
2	15	2025-12-16 08:13:17	batch	1003	Female	No	Yes	No	Month-to-month	Fiber optic	Electronic check	Yes
3	16	2025-12-16 08:13:17	batch	1004	Male	Yes	No	No	One year	DSL	Mailed check	No
4	17	2025-12-16 08:13:17	batch	1005	Female	No	No	No	Month-to-month	Fiber optic	Electronic check	Yes
5	18	2025-12-16 08:13:17	batch	1006	Male	No	Yes	Yes	Two year	DSL	Credit card (automatic)	No
6	19	2025-12-16 08:13:17	batch	1007	Female	Yes	No	No	Month-to-month	Fiber optic	Electronic check	Yes
7	20	2025-12-16 08:13:17	batch	1008	Male	Yes	No	No	One year	DSL	Bank transfer (automatic)	No
8	21	2025-12-16 08:13:17	batch	1009	Female	No	Yes	No	Month-to-month	Fiber optic	Electronic check	Yes
9	22	2025-12-16 08:13:17	batch	1010	Male	Yes	No	No	Two year	DSL	Credit card (automatic)	No

Summary Insights Page:

Select Your Role

Choose your access level:

Business / CRM M...

Prediction & Insights Hub

AI powered decisions based on real data

Select a Page

Batch Prediction

Business Dashboard

Insights & history center

I Help & info

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Single HistoryBatch HistorySummary Insights

Summary Insights — Full DB Analysis

Total Predictions366

Avg Churn Probability51.54%

Negative Feedback %37.98%

Total High Risk Customers171

Churn Decision Breakdown (DB Data)

Sentiment Distribution

Monthly Churn Trend

Customer Segment-wise Analysis

Customer Segment Analysis (Value × New Customer × Churn × Sentiment)

1 Segment Population Count

2 Churn Rate per Segment (%)

3 Avg Churn Probability per Segment

4 Revenue at Risk per Segment

5 Sentiment Distribution per Segment

AI Insights Generator

High churn rate → 46.7%

Negative sentiment → 38.0%

Customer base appears stable.

Download Full Prediction History (DB Data)

Download Full History as CSV

Help & Info Page:

Select Your Role

Choose your access level:

Data Scientist / A...

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Select a Page

Single Prediction

Batch Prediction

Business Dashboard

Explainability (SHAP)

Insights & history center

I Help & info

Model Versions: XGBoost v3 | DL v2 | Last Updated: Nov 2025

Multimodal AI for Telecom Customer Churn & Sentiment Analysis

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Help & Info – Application Guide

Real Business Problem – Why This App?

Telecom companies lose millions every year due to customer churn.

This app helps by: ✓ Predicting who is likely to churn

✓ Identifying why they may leave

✓ Suggesting business retention strategies

This turns data into action → used in CRM & customer retention teams.

Business Value = Reduce churn + Improve satisfaction + retain revenue

Pages & Purpose (Quick Navigation)

Page	Purpose
Single Prediction	Predict churn + sentiment for one customer
Batch Prediction	Upload CSV → Predict multiple customers
Business Dashboard	CLV, segmentation, trends, risk zones
Explainability (SHAP)	Why did the model predict churn?
Insights & History Center	View / filter / download history
I Help & Info	Project info (Interview + Resume Purpose)

Technology Stack (Industry-Level Use Case)

- Machine Learning (XGBoost) → Churn Prediction
- Deep Learning (Neural Network) → Sentiment Analysis
- TF-IDF + MLP → Text Feature Engineering
- SHAP Explainability → Feature Influence (Data Science Standard)
- SQLite Database → Permanent History Storage
- Streamlit → Frontend UI with Role-Based Access
- Domain Use-Case → Telecom | CRM | Customer Retention

Model Performance (Evaluation Metrics)

ML Models – Comparison

	Model	Test Accuracy	ROC-AUC	F1 Score	Remark	Final Score
0	Logistic Regression	0.8573	0.9422	0.9618	✓ Good Fit	0.8871
1	Random Forest	0.9092	0.9522	0.9067	✓ Good Fit	0.9227
2	XGBoost	0.9177	0.9677	0.9177	✓ Best Model	0.9344

Selected Final ML Model: XGBoost

DL Model – Sentiment Analysis

	Model	Test Accuracy	ROC-AUC	F1 Score	Remark	Final Score
0	Neural Network (DL)	0.8815	0	0.8777	✓ Good Fit	0.8796

✓ XGBoost → Final Churn Model

✓ Neural Network → Final Sentiment Model

Both models can be deployed in real telecom CRM systems.

Future Enhancements (Industry Deployment)

- API Deployment using FastAPI / Flask
- Login System with Firebase / OAuth
- Cloud Hosting (AWS / Render / Railway)
- Customer Support Chatbot using GenAI
- CRM Integration with Salesforce / Zendesk / Zoho

Internship Project Details (Resume / LinkedIn)

Field	Details
Project Type	Industrial Internship Project
Company	Infod7T Solutions Pvt. Ltd.
Role	Data Science Intern
Duration	2025
Domain	Telecom – Churn & Customer Retention
Skills Used	ML, DL, NLP, SHAP, SQL, Streamlit

Main Objectives: ✓ Predict customer churn

Understand customer sentiment

Provide business-level insights & retention strategy

About the Developer

Name: Ayesha Banu

Gold Medalist & 1st Rank in M.Sc. Computer Science

Aspiring Data Analyst / ML Engineer / Data Scientist Developer

Email: ayesha28banu@gmail.com

LinkedIn: https://www.linkedin.com/in/ayesha_banu_cs

GitHub: https://github.com/ayesha28banu/telecom_churn_multimodal_ai

Streamlit App Output: Data Scientist / Analyst Role

[Backup / Restore Prediction History Page:](#)

Streamlit App Output Images

➤ This Pages are same as Business / CRM Manager Role pages :

- ❑ Single Prediction Page
- ❑ Batch Prediction Page
- ❑ Business Dashboard Page
- ❑ Insights & history Page
- ❑ Single History Tab
- ❑ Batch History Tab
- ❑ Summary Insights Tab
- ❑ Help & Info page

Select Your Role

Choose your access level:

Data Scientist / A...

Prediction & Insights Hub

AI-powered decisions based on real data

Select a Page

☐ Single Prediction

☐ Batch Prediction

☐ Business Dashboard

☐ Explainability (SHAP)

☒ insights & history center

☐ Help & Info

Model Versions: XGBoost v3 | DL v2.1.1

Get Updated: May 2025

Multimodal AI for Telecom Customer Churn & Sentiment Analysis

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Unified Insights & History Center

View, Analyze & Download Past Predictions — Session + Database

Single History

Batch History

Summary Insights

Backup / Restore

Backup & Restore Prediction History

Backup Current DB to CSV

Upload Old History to Restore DB

Upload CSV History

Drag and drop file here

Limit 200MB per file • CSV

Browse files

Filter DB Records

Filter by Sentiment

All

Filter by Churn

All

	id	timestamp	mode	customer_id	gender	SeniorCitizen	Partner	Dependents	Contract	InternetService	PaymentMethod	Paperless
0	1	2025-12-16 08:00:01	single	10092	Female	Yes	Yes	Yes	Month-to-month	DSL	Electronic check	Yes
1	2	2025-12-16 08:03:07	single	10093	Female	Yes	Yes	No	One year	Fiber optic	Credit card (automatic)	Yes
2	3	2025-12-16 08:06:49	single	1036	Male	No	No	Yes	Month-to-month	No	Electronic check	No
3	4	2025-12-16 08:08:21	single	0001	Male	No	No	Yes	One year	Fiber optic	Bank transfer (automatic)	Yes
4	5	2025-12-16 08:08:39	single	0001	Male	No	No	Yes	One year	Fiber optic	Bank transfer (automatic)	Yes
5	6	2025-12-16 08:09:15	single	0002	Male	No	No	Yes	One year	Fiber optic	Bank transfer (automatic)	Yes
6	7	2025-12-16 08:10:00	single	0002	Male	No	No	Yes	One year	Fiber optic	Bank transfer (automatic)	Yes
7	8	2025-12-16 08:10:28	single	0003	Male	No	No	Yes	Month-to-month	Fiber optic	Bank transfer (automatic)	Yes
8	9	2025-12-16 08:10:40	single	0003	Male	No	No	Yes	Month-to-month	Fiber optic	Bank transfer (automatic)	Yes
9	10	2025-12-16 08:11:01	single	0004	Male	No	No	Yes	Month-to-month	Fiber optic	Bank transfer (automatic)	Yes

Download Filtered DB History

Model Explainability - Shap Insights Page

Single Prediction Explainability Tab:

Select Your Role

Choose your access level:

Data Scientist / A...

Prediction & Insights Hub

AI-powered decisions based on real data

Select a Page

Single Prediction

Batch Prediction

Business Dashboard

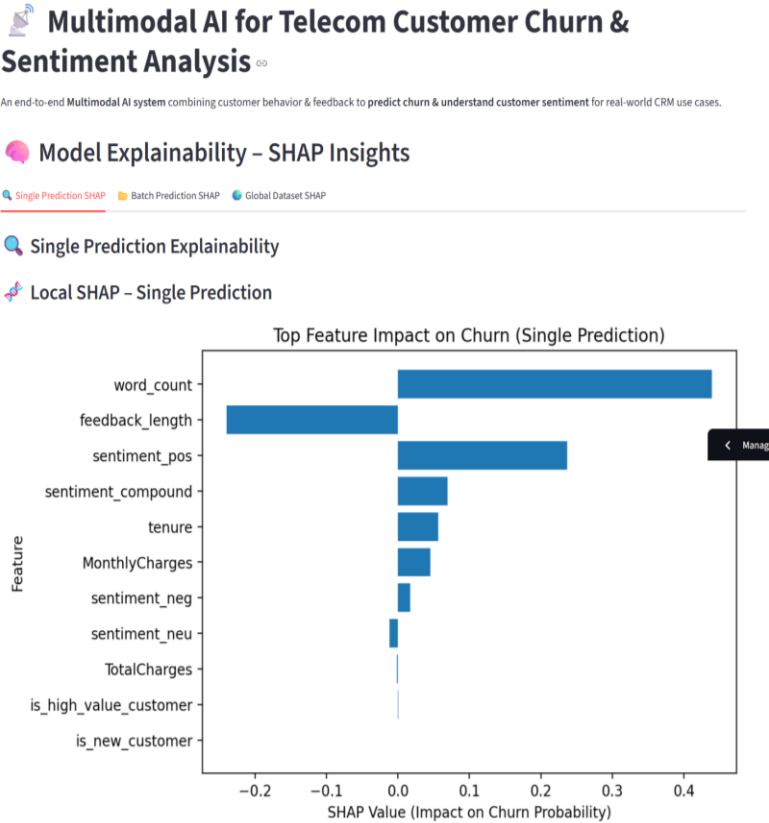
Explainability (SHAP)

insights & history center

Help & Info

Model Versions: XGBoost v3 | DL

v1.1.1 - not final release - Minor Update



Feature Contribution Table

	Feature	Input_Value	SHAP_Contribution
0	word_count	5	0.42
1	feedback_length	36	-0.2401
2	sentiment_pos	0	0.2366
3	sentiment_compound	-0.5859	0.0695
4	tenure	12	0.056
5	MonthlyCharges	700	0.0456
6	sentiment_neg	0.615	0.0172
7	sentiment_neu	0.385	-0.0122
8	TotalCharges	2000	-0.0019
9	is_high_value_customer	0	0.0005

AI Insight:

Most influential feature: word_count — this has the strongest effect on churn.

Local explainability insights generated successfully.

Batch Prediction Explainability Tab :

Select Your Role

Choose your access level:

Data Scientist / A...

Prediction & Insights Hub

AI-powered decisions based on real data

Select a Page

Single Prediction

Batch Prediction

Business Dashboard

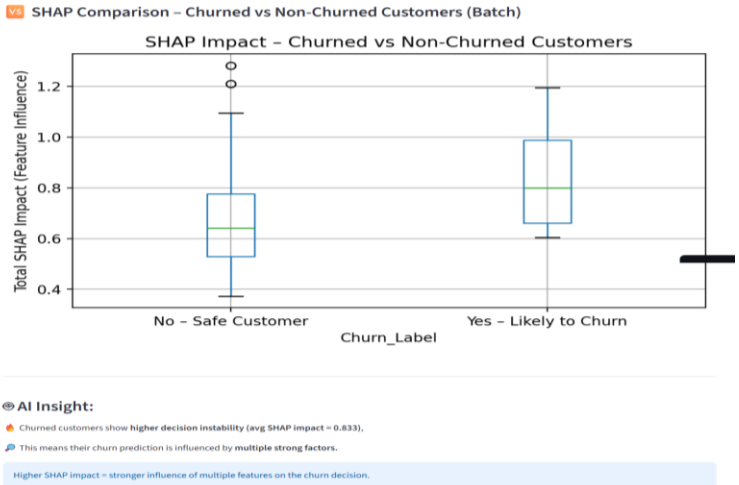
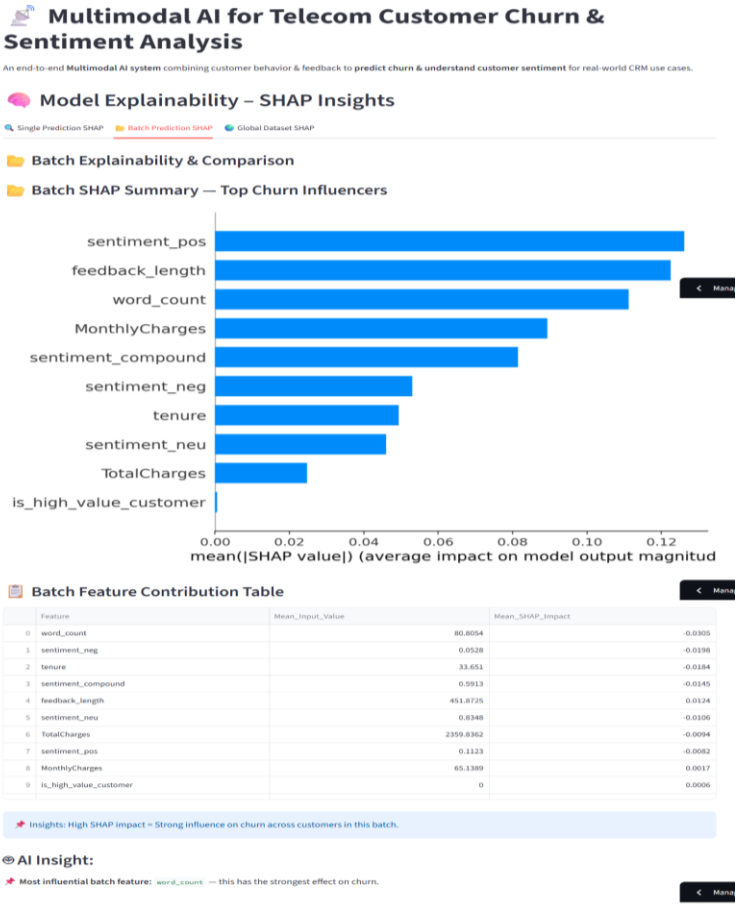
Explainability (SHAP)

insights & history center

Help & Info

Model Versions: XGBoost v3 | DL

v1.1.1 - not final release - Minor Update



Global Explainability Tab :

Select Your Role

Choose your access level:

Data Scientist / A...

Prediction & Insights Hub

AI-powered decisions based on real data

Select a Page

Single Prediction

Batch Prediction

Business Dashboard

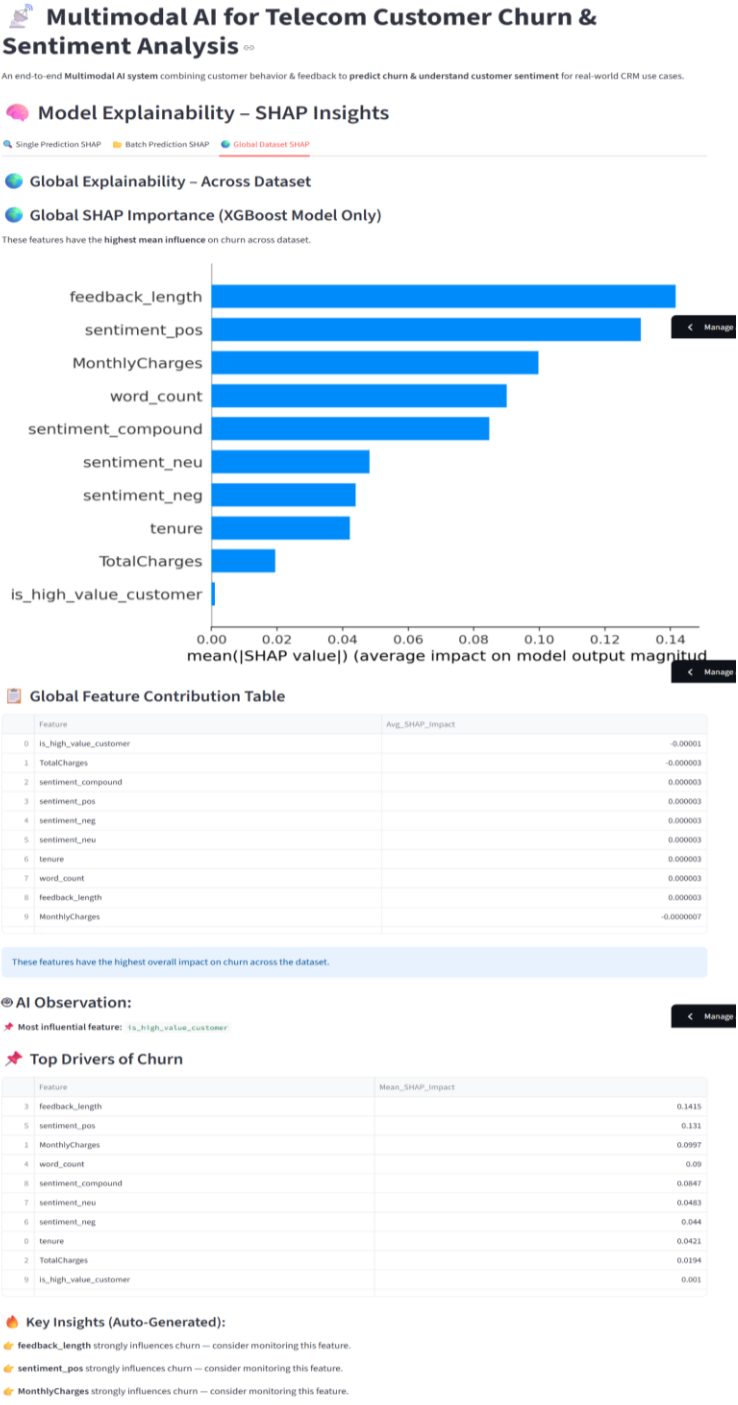
Explainability (SHAP)

insights & history center

Help & Info

Model Versions: XGBoost v3 | DL

v1.1.1 - not final release - Minor Update



Results, Impact & What This Demonstrates

Industry-Grade AI for Telecom

1

Predict WHO Will Leave

XGBoost model with 96.77% ROC-AUC identifies high-risk customers before they churn

2

Understand WHY They Leave

Deep learning sentiment analysis with 88.15% accuracy reveals specific complaint triggers

3

Enable Proactive Retention

Fusion intelligence prioritizes interventions and suggests targeted retention strategies

Technical Skills Demonstrated

End-to-End ML/DL Pipeline

Rare multimodal AI skill combining tabular ML, text DL, and intelligent fusion

Production Deployment

Real-time Streamlit app with SQLite backend, batch processing, and dashboard

Explainable AI

SHAP integration and attention mechanisms for model transparency and trust

Business Impact

CRM-ready outputs with actionable insights and measurable ROI potential

Technology Stack

Languages & Frameworks: Python, TensorFlow, XGBoost, VADER, SHAP

Deployment: Streamlit, SQLite

Techniques: ML, Deep Learning, NLP, Feature Engineering, Model Explainability

Key Takeaway

Multimodal AI = Powerful Customer Intelligence

This deployment-ready pipeline transforms raw telecom data into strategic retention actions, demonstrating production-grade data science capabilities.

Thank You